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Database Management

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Part A:

1:

The screenshot shows a database query editor interface. The top section, titled 'Query', contains an SQL query. The query is as follows:

```
1 SELECT s.studentID, s.studentName
2 FROM student s
3 WHERE s.studentID IN (
4     SELECT e.studentID
5     FROM enrollment e
6     WHERE e.term = 'Spring2026' AND e.courseID = 'CMPT308'
7 )
```

The bottom section, titled 'Data Output', shows the results of the query. It includes a toolbar with icons for various actions and a table of results. The table has two columns: 'studentid' (integer, primary key) and 'studentname' (character varying). The first row of data shows '10' for studentid and 'John Shorts' for studentname.

studentid [PK] integer	studentname character varying
10	John Shorts

2:

Query Query History

```
1 SELECT c.courseID, c.title
2 FROM course c
3 WHERE EXISTS (
4     SELECT 1
5     FROM enrollment e
6     WHERE e.courseID = c.courseID AND e.term = 'Spring2026'
7 )
```

Data Output Messages Notifications

Showing rows: 1 to 7 Page No: 1 of 1

	courseid [PK] character varying	title character varying
1	HIST101	World History
2	CMPT308	Database Manageme...
3	CMPT306	Database Systems
4	CHEM302	Organic Chemistry
5	ECO309	Microeconomics
6	CYBR210	Cyber Security
7	PHY230	Physics I

3:

```
1 SELECT s.studentID, s.studentName
2 FROM student s
3 WHERE NOT EXISTS (
4     SELECT 1
5     FROM enrollment e
6     WHERE e.studentID = s.studentID AND e.term = 'Spring2026'
7 )
8 |
```

Data Output Messages Notifications

Showing rows: 1 to 2 Page No: 1 of 1

	studentid [PK] integer	studentname character varying
1	1	Alice Johnson
2	7	Grace Kim

4:

The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT e.studentID
2 FROM enrollment e
3 WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
4 UNION
5 SELECT e.studentID
6 FROM enrollment e
7 WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'
```

The results pane shows the output of the query, displaying two rows of student IDs. The column is labeled 'studentid' with a data type of 'integer'.

	studentid integer
1	3
2	10

5:

The screenshot shows a SQL IDE interface with a query editor and a results pane. The query editor contains the following SQL code:

```
1 SELECT e.studentID
2 FROM enrollment e
3 WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
4 INTERSECT
5 SELECT e.studentID
6 FROM enrollment e
7 WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'
```

The results pane shows the output of the query, displaying one row of student IDs. The column is labeled 'studentid' with a data type of 'integer'.

	studentid integer
1	3

6:

Query Query History

```

1 SELECT e.studentID
2 FROM enrollment e
3 WHERE e.courseID = 'CMPT308' AND e.term = 'Spring2026'
4 EXCEPT
5 SELECT e.studentID
6 FROM enrollment e
7 WHERE e.courseID = 'CYBR210' AND e.term = 'Spring2026'

```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

studentid	integer
1	10

Part B:

1: $A^+ \Rightarrow \{A\}$, $A \rightarrow B$ so $\{A, B\}$, $B \rightarrow C$ so $\{A, B, C\}$, $AC \rightarrow D$ so $\{A, B, C, D\}$, $D \rightarrow E$ so $\{A, B, C, D, E\} = A^+$

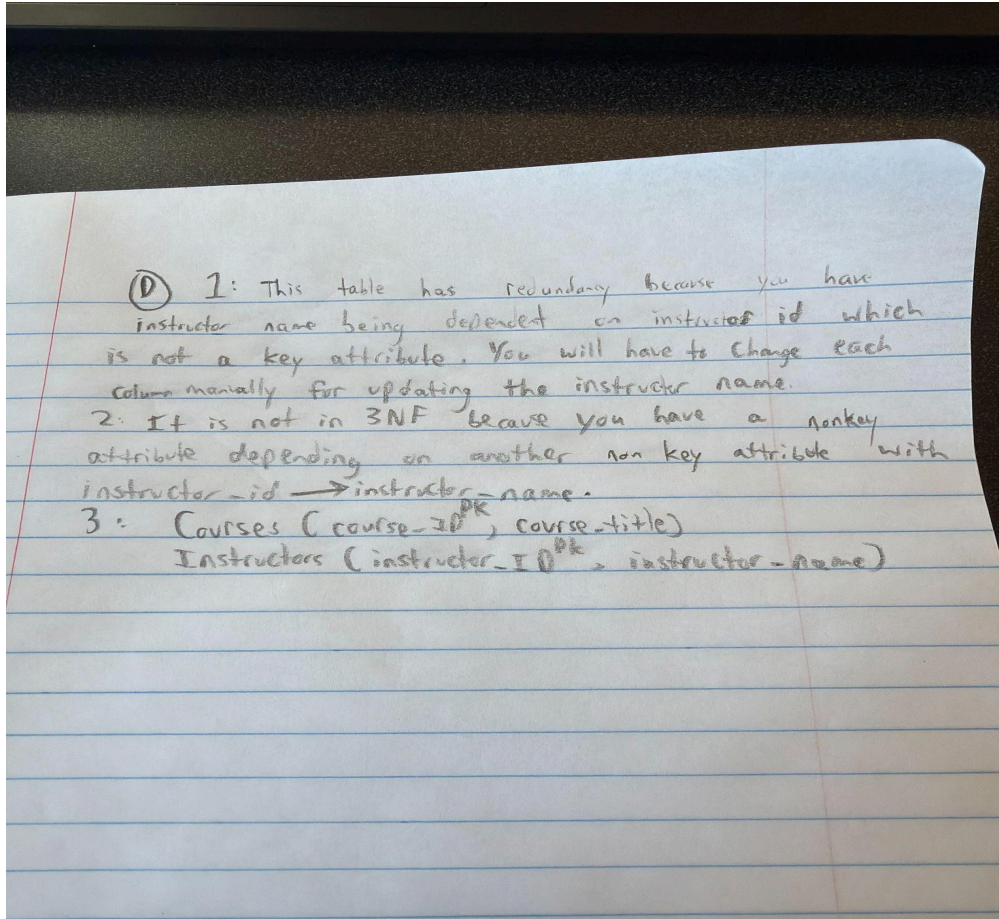
2: $(A, C)^+ \Rightarrow \{A, C\}$, $A \rightarrow B$ so $\{A, B, C\}$, $AC \rightarrow D$ so $\{A, B, C, D\}$, $D \rightarrow E$ so $\{A, B, C, D, E\} = (A, C)^+$

3. The candidate key for R is A and it is a key because A determines everything. $\{A\}$ will confirm B so you will have $\{A, B\}$. $B \rightarrow C$ so $\{A, B, C\}$. A and C $\rightarrow D$ which will be used to get E, $A \rightarrow B \rightarrow C \rightarrow A+C \rightarrow D \rightarrow E$

Part C:

(C) 1: Enrollment wide is in 1NF because all of its attributes are atomic and will have no more than 1 value for 1 enrollment. For example, a student will not have a list of grades for 1 enrollment.
 2: EnrollmentWide does violate 2NF since you have 2 partial dependencies with $\text{student_id} \rightarrow \text{student_name}$ and $\text{course_id} \rightarrow \text{course_title}$.
 3: Students ($\text{student_id}^{\text{pk}}$, student_name)
 (courses ($\text{course_id}^{\text{pk}}$, course_title)
 Enrollments ($\text{student_id}^{\text{pk, fk}}$, $\text{course_id}^{\text{pk, fk}}$, term^{pk} , grade^{pk})

Part D:



GITHUB Link: [DatabaseLabs/Lab5 at main · Not-Patrick1/DatabaseLabs](https://github.com/Not-Patrick1/DatabaseLabs)